

# Tutorial

## Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity.  
Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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## WEEK 6

## 7. THE HYDROGEN ATOM

## • Quantum numbers / Ex. 7.10

For a 3p state give the possible values of  $n$ ,  $\ell$ ,  $m_\ell$ ,  $L$ ,  $L_z$ ,  $L_x$ , and  $L_y$ .

**Solution**

The 3p states corresponds to  $\psi_{31m_\ell}(r, \theta, \phi)$  which has quantum numbrs  $n = 3$  and  $\ell = 1$ , corresponding to the possible values  $m_\ell = \{-\ell, -\ell + 1, \dots, \ell\} = \{-1, 0, 1\}$ .

The solutions of the Hydrogen atom are simultaneous eigenfunctions of  $L^2$  with eigenvalue  $\hbar\ell(\ell + 1)$  and  $L_z$  with eigenvalue  $\hbar m_\ell$ . Therefore  $L^2\psi_{31m_\ell}(r, \theta, \phi) = 2\hbar$ ,  $L\psi_{31m_\ell}(r, \theta, \phi) = \sqrt{2}\hbar$ , and  $L_z\psi_{31m_\ell}(r, \theta, \phi) = m_\ell\hbar$ .

Finally, since  $L_x$  and  $L_y$  do not commute with  $L_z$ , their values cannot be simultaneously determined, but independently  $L_x$  and  $L_y$  can take the same values as  $L_z$  because of symmetry.

## • On the radial probability for the Hydrogen ground state / Examples 7.11, 7.12 and 7.13

Consider the ground state ( $n = 1$ ) of the Hydrogen atom. Calculate (a) the most probable radius for the electron and (b) the average orbital radius. Compare this result with the probability of the electron being at a radius greater than the Bohr radius,  $r > a_0$ .

**Solution**

Due to the spherical symmetry of the (isolated) Hydrogen atom, the general wavefunctions can be decomposed into a radial and an angular parts as

$$\psi_{n\ell m_\ell}(r, \theta, \phi) = R_{n\ell}(r)Y_\ell^{m_\ell}(\theta, \phi).$$

The radial part is normalized as

$$\int_0^\infty dr r^2 R_{n\ell}^*(r) R_{n\ell}(r) = 1,$$

and the angular part is given by the *spherical harmonics* that do *not* depend on the radial coordinate, and are normalized on the unit sphere as

$$\int_0^\pi d\theta \sin(\theta) \int_0^{2\pi} d\phi Y_\ell^{m_\ell*}(\theta, \phi) Y_\ell^{m_\ell}(\theta, \phi) = 1.$$

If we want to calculate the probability that the electron is within a given radius  $r_0$ , we evaluate

$$P(r < r_0) = \int_0^{r_0} dr r^2 \int_0^\pi d\theta \sin(\theta) \int_0^{2\pi} d\phi \psi_{n\ell m_\ell}^*(r, \theta, \phi) \psi_{n\ell m_\ell}(r, \theta, \phi) = \int_0^{r_0} dr r^2 R_{n\ell}^*(r) R_{n\ell}(r),$$

where we carried out the angular integrals using the normalization condition of the spherical harmonics. So we introduce the *radial probability density* as

$$P_{n\ell}(r) = r^2 \int_0^\pi d\theta \sin(\theta) \int_0^{2\pi} d\phi \psi_{n\ell m_\ell}^*(r, \theta, \phi) \psi_{n\ell m_\ell}(r, \theta, \phi) = r^2 R_{n\ell}^*(r) R_{n\ell}(r),$$

which allows us to write the probability of being within a radius as

$$P(r < r_0) = \int_0^{r_0} dr P_{n\ell}(r).$$

Note that the  $r^2$  term in  $P_{n\ell}(r)$  comes from the Jacobian of the integral in spherical coordinates, and expresses the fact that the radial probability density measures the probability of finding the electron in a thin shell of radius  $r$ , the volume of which grows as  $r^2$ .

So the *most probable* radius  $\bar{r}$  is determined by the condition

$$\frac{d}{dr} P_{n\ell}(r) = 0,$$

whereas the *average* radius is given by

$$\langle r \rangle = \int_0^\infty dr r P_{n\ell}(r) = \int_0^\infty dr r^3 R_{n\ell}^*(r) R_{n\ell}(r).$$

The Hydrogen ground state corresponds to quantum numbers  $\{n, m, \ell\} = \{1, 0, 0\}$  (remember that  $0 \leq l < n$  and  $-l \leq m \leq l$ , so for  $n = 1$  this is the only choice), and the total wavefunction is given by  $\psi_{100}(r, \theta, \phi) = R_{10}(r)Y_0^0(\theta, \phi)$ . As  $Y_0^0(\theta, \phi) = 1/(4\pi)$ , the (properly normalized) radial probability density is given by

$$P_{10}(r) = r^2 |R_{10}(r)|^2 = \frac{4r^2}{a_0^3} e^{-2r/a_0},$$

so that

$$0 = \frac{d}{dr} P_{10}(r) = \frac{d}{dr} \frac{4r^2}{a_0^3} e^{-2r/a_0} = \frac{4}{a_0^3} \left( -\frac{2}{a_0} r^2 + 2r \right) e^{-2r/a_0}$$

yielding

$$\bar{r} = a_0.$$

Instead, the average value is given by

$$\langle r \rangle = \frac{4}{a_0^3} \int_0^\infty dr r^3 e^{-2r/a_0} = \frac{3}{2} a_0.$$

Finally, calculating the probability for  $r > a_0$  yields

$$P(r > a_0) = \frac{4}{a_0^3} \int_{a_0}^\infty dr r^3 e^{-2r/a_0} = 5e^{-2} \approx 0.68,$$

meaning that there is about 68 % probability for the electron to be found at a distance greater than the Bohr radius  $a_0$ , which is in line with the result obtained for the expectation value  $\langle r \rangle$ .

In both cases, the result of the indefinite integral is given by the following formula:

$$\int dx x^m e^{bx} = e^{bx} \sum_{k=0}^m (-1)^k \frac{m! x^{m-k}}{(m-k)! b^{k+1}}.$$

#### • Hydrogen-like ions / Ex. 7.46

Consider a hydrogen-like atom such as  $\text{He}^+$  or  $\text{Li}^{++}$  that has a single electron outside a nucleus of charge  $q = Ze$ . (a) Rewrite the Schrödinger equation with the new Coulomb potential. (b) What change does this new potential have on the separation of variables? (c) Will the radial wave functions be affected? Explain. (d) Will the spherical harmonics be affected? Explain.

#### Solution

(a) Only the potential term changes, the Schrödinger equation becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2\mu}{\hbar^2} \left( E + \frac{Ze^2}{4\pi\epsilon_0 r} \right) \psi = 0.$$

(b) The equation remains separable, only the radial part changes.

(c) The radial wavefunctions will change, but their functional forms will remain the same with  $r$  rescaled. In particular, the Bohr radius  $a_0$  is replaced by

$$a_Z = \frac{4\pi\epsilon_0 \hbar^2}{\mu Z e^2}.$$

(d) No, the  $\theta$ - and  $\phi$ -dependent parts remain the same since the potential maintains its spherical symmetry.

• **2p orbitals of the hydrogen atom**

The 2p orbitals ( $n = 2$ ,  $l = 1$ ) of hydrogen are given in the standard angular momentum  $\psi_{nlm}$  basis by

$$\begin{aligned}\psi_{210}(r, \theta, \phi) &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \cos \theta, \\ \psi_{21\pm 1}(r, \theta, \phi) &= \frac{1}{8\sqrt{\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \sin \theta e^{\pm i\phi}.\end{aligned}$$

It is also customary to use the Cartesian basis functions

$$\begin{aligned}\psi_{2p_x}(x, y, z) &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{x}{a_0} e^{-\frac{r}{2a_0}}, \\ \psi_{2p_y}(x, y, z) &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{y}{a_0} e^{-\frac{r}{2a_0}}, \\ \psi_{2p_z}(x, y, z) &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{z}{a_0} e^{-\frac{r}{2a_0}},\end{aligned}$$

where  $r = \sqrt{x^2 + y^2 + z^2}$ .

Express the Cartesian basis functions as linear combinations of  $\psi_{21m}$ . Express the inverse transformation as well, i.e. the  $\psi_{21m}$  basis functions as linear combinations of the Cartesian basis functions.

**Solution**

We use the transformation between Cartesian and polar coordinates:

$$\begin{aligned}x &= r \sin \theta \cos \phi, \\ y &= r \sin \theta \sin \phi, \\ z &= r \cos \theta.\end{aligned}$$

Substituting this into the Cartesian functions (i.e., writing them in polar coordinates) we immediately find that  $\psi_{2p_z} = \psi_{210}$ . For the other two basis functions, we use the exponential form of the trigonometric functions:

$$\begin{aligned}\psi_{2p_x} &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \sin \theta \cos \phi = \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \sin \theta \frac{e^{i\phi} + e^{-i\phi}}{2} = \frac{1}{\sqrt{2}} (\psi_{211} + \psi_{21-1}), \\ \psi_{2p_y} &= \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \sin \theta \sin \phi = \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \sin \theta \frac{e^{i\phi} - e^{-i\phi}}{2i} = \frac{1}{\sqrt{2}i} (\psi_{211} - \psi_{21-1}).\end{aligned}$$

To find the inverse transformation, we only need to solve a system of linear equations:

$$\begin{aligned}\psi_{210} &= \psi_{2p_z}, \\ \psi_{211} &= \frac{1}{\sqrt{2}} (\psi_{2p_x} + i\psi_{2p_y}), \\ \psi_{21-1} &= \frac{1}{\sqrt{2}} (\psi_{2p_x} - i\psi_{2p_y}).\end{aligned}$$

• **Orthogonality of hydrogen wavefunctions**

We define the *inner product* or *overlap* of two one-dimensional wavefunctions  $\phi$  and  $\psi$  as

$$\langle \phi | \psi \rangle = \int dx \phi^*(x) \psi(x),$$

where the integral runs over the entire space where the particles can appear. In three dimensions the integral is replaced with a volume integral. This expression is analogous to the norm of the wavefunction, which is the same as the inner product of the wavefunction with itself.

We say that two wavefunctions are *orthogonal* if their overlap is zero. Check that the hydrogen  $2p$  orbitals  $\psi_{21m}$  for  $m = -1, 0, 1$  form an orthonormal set, i.e. they are mutually orthogonal and the norm of each one is 1:

$$\langle \psi_{21m} | \psi_{21m'} \rangle = \delta_{mm'}.$$

Be careful to use the correct volume integral in polar coordinates.

Using this result, also check that the Cartesian  $2p$  wavefunctions from the previous problem also form an orthonormal set. Using the bilinearity of the inner product you can prove this without evaluating any integrals.

**Solution**

The normalization of  $\psi_{211}$  is shown in example 7.2 of the book, the other normalization conditions can be checked similarly.

Let us check the overlap

$$\begin{aligned} \langle \psi_{210} | \psi_{211} \rangle &= \int_0^\infty dr r^2 \int_0^\pi d\theta \sin(\theta) \int_0^{2\pi} d\phi \psi_{210}^*(r, \theta, \phi) \psi_{211}(r, \theta, \phi) \\ &= \frac{1}{64\pi a_0^3} \int_0^\infty \frac{r^4}{a_0^2} e^{-\frac{r}{a_0}} dr \int_0^\pi \cos \theta \sin \theta d\theta \int_0^{2\pi} e^{i\phi} d\phi, \end{aligned}$$

where we separated the integral. Let us evaluate the last integral:

$$\int_0^{2\pi} e^{i\phi} d\phi = \left[ \frac{e^{i\phi}}{i} \right]_0^{2\pi} = 0,$$

hence the overlap vanishes. The other overlap integrals are evaluated similarly.

The bilinearity of the inner product means that the following identities are true for any functions  $\psi_i$ , and complex numbers  $a, b$ :

$$\begin{aligned} \langle \psi_1 | a\psi_2 + b\psi_3 \rangle &= a\langle \psi_1 | \psi_2 \rangle + b\langle \psi_1 | \psi_3 \rangle, \\ \langle a\psi_1 + b\psi_2 | \psi_3 \rangle &= a^*\langle \psi_1 | \psi_3 \rangle + b^*\langle \psi_2 | \psi_3 \rangle. \end{aligned}$$

Let us check the normalization of the  $\psi_{2p_x}$  wavefunction:

$$\begin{aligned} \langle \psi_{2p_x} | \psi_{2p_x} \rangle &= \frac{1}{2} \langle \psi_{211} + \psi_{21-1} | \psi_{211} + \psi_{21-1} \rangle \\ &= \frac{1}{2} (\langle \psi_{211} | \psi_{211} \rangle + \langle \psi_{211} | \psi_{21-1} \rangle + \langle \psi_{21-1} | \psi_{211} \rangle + \langle \psi_{21-1} | \psi_{21-1} \rangle) \\ &= \frac{1}{2} (1 + 0 + 0 + 1) = 1. \end{aligned}$$

We can also evaluate the overlaps, for example:

$$\begin{aligned} \langle \psi_{2p_y} | \psi_{2p_x} \rangle &= -\frac{1}{2i} \langle \psi_{211} - \psi_{21-1} | \psi_{211} + \psi_{21-1} \rangle \\ &= -\frac{1}{2i} (\langle \psi_{211} | \psi_{211} \rangle + \langle \psi_{211} | \psi_{21-1} \rangle - \langle \psi_{21-1} | \psi_{211} \rangle - \langle \psi_{21-1} | \psi_{21-1} \rangle) \\ &= -\frac{1}{2i} (1 + 0 - 0 - 1) = 0. \end{aligned}$$

The other overlaps can be computed similarly.

• **Angular momentum commutation relations**

The angular momentum operators in quantum mechanics are defined analogously to classical mechanics by  $\mathbf{L} = \mathbf{r} \times \mathbf{p}$  using the position and momentum operators. The various angular momentum components obey the commutation relations

$$[L_\alpha, L_\beta] = i\hbar\epsilon^{\alpha\beta\gamma}L_\gamma,$$

where we used the commutator  $[A, B] = AB - BA$ , and  $\epsilon^{\alpha\beta\gamma}$  is Levi-Civita's completely-antisymmetric tensor, defined so that  $\epsilon^{\alpha\beta\gamma} = \pm 1$  for every *even/odd* permutation of  $\{x, y, z\}$  and  $\epsilon_{\alpha\beta\gamma} = 0$  if any index is repeated. Using the Heisenberg commutation relation for position and momentum  $[r_\alpha, p_\beta] = i\hbar\delta_{\alpha\beta}$  (using the Kronecker delta:  $\delta_{\alpha\beta} = 1$  if  $\alpha = \beta$ ,  $\delta_{\alpha\beta} = 0$  if  $\alpha \neq \beta$ ), confirm the commutator relation

$$[L_z, L_x] = i\hbar L_y.$$

Analogously check that

$$[L_x, L_y] = i\hbar L_z.$$

and

$$[L_y, L_z] = i\hbar L_x.$$

Using the angular momentum commutation relations, check that  $\mathbf{L}^2 = \sum_{\alpha=x,y,z} L_\alpha^2$  commutes with all the angular momentum components:

$$[\mathbf{L}^2, L_\alpha] = 0.$$

**Solution**

In order to derive these kinds of relations, a possible strategy is to substitute the expression of the angular momentum in terms of position and momentum operators as

$$[L_z, L_x] = [xp_y - yp_x, yp_z - zp_y] = [xp_y, yp_z - zp_y] - [yp_x, yp_z - zp_y] = [xp_y, yp_z] - [xp_y, zp_y] - [yp_x, yp_z] + [yp_x, zp_y].$$

Of all the above commutators, only two terms survive, so that

$$[L_z, L_x] = i\hbar(p_x z - xp_z) = L_y.$$

**Strategy**

In order to obtain the final result, one can exploit the following properties of commutators

$$[A, BC] = B[A, C] + [A, B]C \quad (1)$$

and

$$[AB, C] = A[B, C] + [A, C]B \quad (2)$$

Consider, e.g., the first commutator,  $[xp_y, yp_z]$ , and apply property (1) with

$$xp_y = A \quad \text{and} \quad yp_z = BC$$

so that

$$[xp_y, yp_z] = y[xp_y, p_z] + [xp_y, y]p_z,$$

and then apply property (2) with

$$[AB, C] = B[A, C] + [A, C]B$$

with

$$xp_y = AB \quad \text{and} \quad p_z = C \quad \text{or} \quad y = C$$

respectively for each of the terms. Finally (being careful not to interchange the order of the operators) the commutator is decomposed into the following

$$[xp_y, yp_z] = yx[p_y, p_z] + y[x, p_z]p_y + x[p_y, y]p_z + [x, y]p_y p_z$$

and according to the commutation relation for position and momentum operators

$$[r_\alpha, p_\beta] = i\hbar\delta_{\alpha\beta}, \quad \alpha, \beta \in \{x, y, z\}$$

the sole non-zero term, given by

$$x[p_y, y]p_z = -i\hbar xp_z.$$

The same procedure is applied to all other commutators, and the sole other non-zero term originates from the last one, yielding

$$[yp_x, zp_y] = i\hbar p_x z$$

thus

$$[L_z, L_x] = i\hbar(p_x z - xp_z) = i\hbar L_y.$$

All other commutation relations can be demonstrated analogously.

For the commutation relation with  $\mathbf{L}^2$ , we expand the commutator for  $\alpha = x$ :

$$[\mathbf{L}^2, L_x] = [L_x^2, L_x] + [L_y^2, L_x] + [L_z^2, L_x].$$

The first term vanishes, as every operator commutes with itself and its powers. Let us expand the second term using the commutation relation  $[L_x, L_y] = i\hbar L_z$ , implying  $L_x L_y = i\hbar L_z + L_y L_x$ :

$$[L_y^2, L_x] = L_y^2 L_x - L_x L_y^2 = L_y^2 L_x - (i\hbar L_z + L_y L_x) L_y = L_y^2 L_x - i\hbar L_z L_y - L_y (i\hbar L_z + L_y L_x) = -i\hbar (L_z L_y + L_y L_z).$$

Similarly expanding the third term we find

$$[L_z^2, L_x] = i\hbar (L_z L_y + L_y L_z),$$

hence these terms cancel, proving the commutation relation, the other components can be computed analogously.